

Second Exams Spring 2019/2020

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$s_n = \ln\left(\frac{3n-1}{n+5}\right) + \sin n$

- ☒ It is divergent ✓
- ☐ It is convergent with sum $\ln 3$
- ☐ It is convergent with sum $-\ln 5$
- ☐ It is convergent with sum ∞

2. (5 Points)

If the n -th partial sum of the series $\sum_{n=1}^{\infty} a_n$ is $s_n = \frac{2n-1}{n+2}$, find a_n .

- ☒ $a_1 = \frac{1}{3}, a_n = \frac{5}{(n+1)(n+2)} \quad (n \geq 2)$ ✓
- ☐ $a_1 = \frac{1}{3}, a_n = \frac{5}{(n+2)(n+3)} \quad (n \geq 2)$
- ☐ $a_1 = \frac{3}{2}, a_n = \frac{5}{(n+1)(n+2)} \quad (n \geq 2)$
- ☐ $a_1 = \frac{3}{2}, a_n = \frac{5}{(n+2)(n+3)} \quad (n \geq 2)$

3. Which of the following is correct about the series

(5 Points)

$$\sum_{n=1}^{\infty} \left(\frac{2}{3^n} + \arctan n \right)$$

- ☒ It is divergent ✓
- ☐ It is convergent with sum 1
- ☐ It is convergent with sum $1 + \frac{\pi}{2}$
- ☐ It is convergent with sum $\frac{\pi}{2}$

4. What is the sum of the series

(5 Points)

$$\sum_{n=1}^{\infty} \left(\frac{5}{3^n} + \frac{(-2)^n}{5^{n-1}} \right)$$

- ☒ $\frac{15}{14}$ ✓
- ☐ $\frac{7}{16}$
- ☐ $\frac{12}{5}$
- ☐ $\frac{43}{15}$

5. Which of the following is true for the series

(4 Points)

$$\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n+2}\right)$$

- ☒ It is divergent ✓
- ☐ It is convergent with sum 0
- ☐ It is convergent with sum $\ln 3 - \ln 2$
- ☐ It is convergent with sum $\frac{1}{2}$

6. Which of the following is correct for the series

(4 Points)

$$\sum_{n=2}^{\infty} \frac{1}{(n-2)^2 + 1}$$

- ☒ The Integral Test applies and concludes that the series is convergent. ✓
- ☐ $f(x) = \frac{1}{(x-2)^2 + 1}$ is not decreasing on $[2, \infty)$, so The Integral Test does not apply.
- ☐ $f(x) = \frac{1}{(x-2)^2 + 1}$ is not positive on $[2, \infty)$, so The Integral Test does not apply.
- ☐ The Integral Test applies and concludes that the series is divergent.

7. Find all positive values of c for which the following series converges.

(5 Points)

$$\sum_{n=1}^{\infty} \frac{1+n^2}{n^{c-1}}$$

- ☒ $c > 4$ ✓
- ☐ $c > 2$
- ☐ $c \geq 4$
- ☐ $c \geq 2$

8. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \sin(e^{-n})$$

- ☒ It is convergent by the Limit Comparison Test with a geometric series. ✓
- ☐ It is divergent by the Limit Comparison Test with a geometric series.
- ☐ It is divergent by the Integral Test.
- ☐ It is convergent by the Limit Comparison Test with a p -series.

9. Is the following series convergent or divergent?

(4 Points)

$$\sum_{n=1}^{\infty} \frac{n^2 + 10^n}{n^{10} + 2^n}$$

- ☒ Divergent ✓
- ☐ Convergent

10. Is the following series convergent or divergent?

(4 Points)

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n^2)}$$

- ☒ Convergent ✓
- ☐ Divergent

11. Which of the following series is divergent given that

(5 Points)

$$b_n > 0 \text{ and } \sum_{n=1}^{\infty} b_n = 3$$

- ☒ $\sum_{n=1}^{\infty} (b_n)^{-2}$ ✓
- ☐ $\sum_{n=1}^{\infty} 5b_n$
- ☐ $\sum_{n=1}^{\infty} (b_n)^3$
- ☐ $\sum_{n=1}^{\infty} \frac{b_n}{n}$

12. Which of the following is correct about the series

(4 Points)

$$\sum_{n=1}^{\infty} a_n$$

- ☒ If it is absolutely convergent, then it is convergent. ✓
- ☐ If $a_n > 0$, then the series is absolutely convergent.
- ☐ If it is not absolutely convergent, then it is conditionally convergent.

13. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} \left(\frac{n-1}{n+1} \right)^{n^2}$$

- ☒ It is convergent by the Root Test. ✓
- ☐ It is divergent by the Test for divergence.
- ☐ It is divergent by the Root Test.
- ☐ It is convergent by the Alternating Series Test.

14. Determine which of the series below is conditionally convergent.

(5 Points)

- ☒ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{\sqrt[k]{k}}$ ✓
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^k \arctan k}{k^4}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^k k^3}{3^k}$
- ☐ $\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^{-2}}$

15. Which of the following is correct for the series

(5 Points)

$$\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n^2}\right)$$

- ☒ It is absolutely convergent. ✓
- ☐ It is conditionally convergent.
- ☐ It is divergent.
- ☐ None of these.

16. The interval of convergence of a power series is $[1, 5]$. What is the center of this power series?

(5 Points)

- ☒ 3 ✓
- ☐ 2
- ☐ -1
- ☐ It also depends on the coefficients of the power series.

17. Find the radius of convergence of the power series.

(5 Points)

$$\sum_{n=0}^{\infty} \frac{(2x-1)^{3n}}{(3n)!}$$

- ☒ ∞ ✓
- ☐ $\frac{3}{2}$
- ☐ $\frac{2}{3}$
- ☐ 1

18. Find the interval of convergence of the power series

(5 Points)

$$\sum_{n=1}^{\infty} \frac{(x-3)^n}{n^2 2^n}$$

- ☒ $[1, 5]$ ✓
- ☐ $[1, 5)$
- ☐ $(2, 4]$
- ☐ $[2, 4]$

19. Use the power series representation centered at 0 of $f(x)$ to find the sum s .

(5 Points)

$$f(x) = \frac{1}{(1+x)^2}, \quad s = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{2^n}.$$

- ☒ $s = \frac{2}{9}$ ✓
- ☐ $s = \frac{4}{9}$
- ☐ $s = \frac{2}{3}$
- ☐ $s = \frac{4}{3}$

20. Which of the following is a power series representation of

(5 Points)

$$f(x) = \frac{1}{2-x}$$

- ☒ $\sum_{n=0}^{\infty} (x-1)^n, \quad x \in (0, 2)$ ✓
- ☐ $\sum_{n=0}^{\infty} \frac{2^n x^n}{n+1}, \quad x \in (-\frac{1}{2}, \frac{1}{2})$
- ☐ $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}}, \quad x \in (-2, 2)$

21. Evaluate the indefinite integral as a power series.

(5 Points)

$$\int \left(\sum_{n=0}^{\infty} \frac{(n+1)x^{2n+1}}{n!} \right) dx$$

- ☒ $C + \sum_{n=0}^{\infty} \frac{x^{2n+2}}{2 \cdot n!}$ ✓
- ☐ $C + \sum_{n=0}^{\infty} \frac{(n+1)x^{2n+2}}{(2n+1)n!}$
- ☐ $C + \sum_{n=0}^{\infty} \frac{(n+1)(2n+1)x^{2n}}{n!}$